Appendix A: Detailed Methodology

A.1 Data Sources and Collection

The diagnostic integrates quantitative and qualitative inputs from multiple organizational layers to ensure analytical depth, data integrity, and traceability. Each stream captures a distinct dimension of the analyzed workflows.

Technical Feasibility Assessments (TFA). Completed by IT and system owners to document system environment, data quality, integration feasibility, and security posture. TFAs establish the technical conditions that determine whether automation is feasible within each workflow.

Process Insights Forms (PIF). Completed by process owners and end users to capture operational realities. PIFs measure process clarity, change magnitude, and workflow brittleness, and collect metrics such as process volume, frequency, average duration, and observed error rates. PIF inputs feed readiness scoring and ROI modeling.

Stakeholder Interviews. Sessions with department leads and subject-matter experts validate automation candidates, contextualize TFA and PIF responses, and provide data not found in system documentation, including salary ranges, role counts per workflow, and high-level process maps.

End-User Interviews. Discussions with those performing the work verify PIF inputs, enable detailed workflow mapping, surface edge cases, and isolate high-friction or error-prone stages.

Documentation Review. Review of workflow diagrams, SOPs, and system documentation to cross-check form data and confirm process conditions. Where available, policy documents, screen captures, or task logs are incorporated as supporting evidence.

For a complete record of data inputs, see **Appendix D: Data Provenance and Audit Trail**. It lists, per workflow, the data type, source, respondent name and role, department, collection method, and date.

A.2 Diagnostic Frameworks

Automat-IA applies a structured, multi-layered framework that translates technical and operational inputs into measurable readiness and financial performance indicators.

Technical Feasibility Assessment (TFA).
21 weighted questions scored 1 to 5, with a “Don’t know” option to flag gaps. Section weights:

System Environment 25 percent, Data Quality 30 percent, Integration Feasibility 30 percent, Security and Compliance 15 percent. TFA outputs a 0 to 100 technical readiness score.

Process Insights Form (PIF).
40 questions capturing operational readiness across process clarity, change magnitude, and workflow brittleness, plus quantitative metrics for volumes, times, participants, and quality. When multiple respondents exist, responses are aggregated with variance tracking.

Readiness Rubric.
Combined readiness score per workflow or opportunity using TFA 60 percent and PIF 40 percent, normalized 0 to 100. This score represents estimated implementation effort and feeds the Opportunity Matrix vertical coordinate for effort.

Automat-IA ROI Calculator.
Quantifies financial impact using PIF operational data and cost assumptions from the diagnostic. The calculator applies a double-layer conservatism model, producing annual benefit, payback period, and related business metrics. All calculations are version-controlled and auditable.

Opportunity Matrix.
Final analytical layer that unifies readiness and ROI into a single prioritization framework.

- X axis: Business Impact, a normalized score derived from efficiency and value.
- Y axis: Implementation Effort, represented by the Readiness Rubric score.

A.3 Data Integrity and Knowledge Gap Handling

Gap capture in TFAs. Every question supports a “Don’t know” response that records reasons and potential sources. Automat-IA then: 1) extracts the gaps, 2) assigns follow-ups, 3) updates responses once verified, and 4) records the contributor.

Midpoint response rule. If no stakeholder can provide a verified answer, a neutral midpoint score of 3 on the 1 to 5 scale is applied to preserve dataset completeness. Each midpoint substitution is explicitly logged for auditability.

A.4 Business Impact Scoring Methodology

Business Impact expresses expected value relative to company scale for fair comparison across clients.

Base Impact components and normalization.

- ROI Score: (Annualrun−rateROI÷Companyannualrevenue)×1000(Annual run-rate ROI ÷ Company annual revenue) × 1000(Annualrun−rateROI÷Companyannualrevenue)×1000, capped at 100.

- Hours Score: (FTEsavings÷Totalemployees)×1000(FTE savings ÷ Total employees) × 1000(FTEsavings÷Totalemployees)×1000, capped at 100.

- **Base Impact (0 to 100)** equals ROI Score 60 percent plus Hours Score 40 percent.

Normalization by revenue and FTEs prevents bias toward larger firms and captures the relative significance of benefits.

A.5 ROI Modeling Methodology

Double-layer conservatism.

Layer 1: Rubric-based base estimation.
Structured rubrics convert PIF inputs into base percentages for time savings, quality improvement, and revenue reallocation, considering process type, complexity, repetition, and data structure.

Layer 2: Scenario multipliers.
Risk-adjusted multipliers reflect adoption rates, exception handling, coverage, and integration maturity. Conservative scenario applies downward adjustments across benefit categories.

Key formulas.

- Annual Benefit = Time savings value + Quality improvement value + Revenue uplift value.

- Payback (months) = Implementation cost ÷ Annual Benefit × 12.

- ROI percent (used in methodology and matrices, even if not shown in every table) = (AnnualBenefit−Annualcost)÷Annualcost(Annual Benefit − Annual cost) ÷ Annual cost(AnnualBenefit−Annualcost)÷Annualcost × 100.

Inputs, parameters, and versions are documented in **Appendix E: ROI Calculator Methodology**.

A.6 Opportunity Matrix Methodology

The Opportunity Matrix converts readiness and business impact into a transparent prioritization model.

Two-matrix framework.

1. **Base Opportunity Matrix 0 to 100.** Pure efficiency comparison on a consistent company-relative basis using Base Impact.

2. **Enhanced Opportunity Matrix 0 to 150.** Adds value beyond efficiency:

    - Error Reduction Bonus up to 25 points based on measured error rates and rework avoided.

    - Revenue Uplift Bonus up to 25 points based on validated capacity reallocation toward revenue activities.
      Enhanced Impact = Base Impact + Error Bonus + Revenue Bonus. Enhanced is always greater than or equal to Base.

Axes.

- X axis: Business Impact (Base or Enhanced, depending on the view presented).

- Y axis: Implementation Effort (Readiness Rubric 0 to 100).

Quadrants.

- Quick Wins: High impact and low effort.

- Strategic Bets: High impact and high effort.

- Nice-to-Haves: Low impact and low effort.

- Deprioritize: Low impact and high effort.

A.7 Roadmap Development Methodology

Roadmap sequencing is derived directly from matrix positions and diagnostic evidence.

Phase 1 Quick Wins 0 to 6 months. Low effort and high impact items executed first to deliver ROI quickly and validate governance, data connections, and monitoring.

Phase 2 Foundational Enhancements 6 to 12 months. Address data quality, integration, and documentation gaps and pilot mid-complexity automations to raise readiness.

Phase 3 Strategic Implementations 12 to 18 plus months. High-impact, higher-effort automations that extend across systems and functions, often with AI components and orchestration requirements.

Sequencing also considers technical dependencies, value realization timing, resource capacity, change management cadence, and budget cycles.

A.8 Methodological Limitations

This diagnostic is a point-in-time assessment of readiness and potential value, not a detailed implementation plan.

Reliance on client data. TFAs, PIFs, and interviews are self-reported. Inaccuracies or omissions may affect precision. Midpoint rules and follow-ups mitigate but do not eliminate this risk.

Scope boundary. Findings focus on feasibility, maturity, and impact at diagnostic depth. Solution design, integration validation, and vendor selection occur in subsequent phases.

Uncertainty and change. System changes, policy shifts, or market conditions may alter feasibility and benefits. Conservative modeling and normalization are used to avoid overstating impact.
